Cross-Reactivity Specificity Matrix

Task 6 • Dry Lab Deliverable

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A multiplexed electrochemical biosensor requires that each substrate motif is cleaved selectively by its target protease and not by any other panel member. This report presents a quantitative cross-reactivity specificity matrix for the six-protease diagnostic panel, computed from position-specific scoring matrices (PSSMs) built from all confirmed cleavage events in the MEROPS substrate database. Panel motifs for MMP-8, MMP-9, NE, and Fecal are the top-ranked candidates from the Task 7 ML substrate predictor; MMP14 uses the validated electrochemical sensor motif; FAP uses the Edosada literature reference. Of 30 off-diagonal pairs, 23 are orthogonal (score < 0.05) and 7 are flagged (≥ 0.20).

Task 6: Cross-Reactivity Specificity Matrix

Data Source and PSSM Construction

Confirmed cleavage-site sequences were downloaded from the MEROPS bulk release (`Substrate_search.txt`, accessed July 2026; 108,183 entries total). Records were filtered to the six MEROPS identifiers in Table 1. For each of the eight P4–P4' positions, amino acid frequencies were computed with a Laplace pseudocount of 0.5 and converted to $\log_2$ odds against Swiss-Prot background frequencies (Robinson & Robinson 1991), yielding one 8×20 PSSM per protease. The fecal serine protease PSSM pools three sub-entries (trypsin-1, chymotrypsin B, elastase-1); it is dominated by trypsin-1 (18,131 of 18,251 entries), which confers strong Arg/Lys specificity at P1. FAP has only 2 unique cleavage windows in MEROPS: its PSSM is statistically non-informative and all FAP-related values are flagged throughout.

Protease	MEROPS ID	Unique windows	Confidence	Notes
MMP14	M10.014	129	medium	MT1-MMP
MMP-8	M10.002	96	medium	Neutrophil collagenase
MMP-9	M10.004	341	high	Gelatinase B
NE	S01.131	327	high	Neutrophil elastase
Fecal	S01.151/2/3	18,251	high	Trypsin-1 dominated
FAP	S09.007	2	**unusable**	See §1.5

Table 1: MEROPS training data per protease. Unique windows after 3-letter-to-1-letter conversion, standard-amino-acid filtering, and deduplication.

Panel Motifs

Table 2 lists the six motifs evaluated as column vectors. MMP-8, MMP-9, NE, and Fecal use the top-ranked candidates from the Task 7 ML substrate predictor (this work). MMP14 retains the experimentally validated sensor motif SGRSENIR. FAP uses the Edosada literature reference because the Task 7 FAP model is degenerate ($n = 2$ positives).

Protease	Motif	Source	Status
MMP14	`SGRSENIR`	Ohuchi et al. 1997	Fixed, validated
MMP-8	`KSQFLQGT`	Task 7 ML prediction	LR AUROC 0.773, $S = 2.02$
MMP-9	`GAPPVQAP`	Task 7 ML prediction	LR AUROC 0.774, $S = 2.39$
NE	`PETISLPP`	Task 7 ML prediction	LR AUROC 0.825, $S = 17.9$
Fecal	`DLPRALDQ`	Task 7 ML prediction	LR AUROC 0.938, $S = 1.98$
FAP	`GPRFPIGS`	Edosada et al. 2006	Literature; ML unreliable

Table 2: Panel motifs used as column vectors in the specificity matrix. S = ML selectivity index (target probability / maximum competitor probability) from Task 7.

Specificity Matrix

Each cell in Table 3 is the PSSM $\log_2$-odds score of the column motif under the row protease's matrix, divided by the absolute self-score (so the diagonal reference is ±1.00). Values ≥ 0.20 are **flagged**; values < 0.05 in absolute magnitude are considered orthogonal.

Protease	MMP14	MMP-8	MMP-9	NE	Fecal	FAP
MMP14	−1.00†	**0.86**	**0.54**	−1.72	−0.12	−0.76
MMP-8	−0.41	1.00	**0.70**	−0.89	−0.38	**0.52**
MMP-9	−0.59	**0.48**	1.00	−0.62	−0.49	**0.64**
NE	−0.82	−0.39	−0.11	1.00	−0.10	−0.74
Fecal	−1.64	−0.90	−1.08	−0.98	1.00	−1.91
FAP	−1.15	**0.42**	0.02	−1.20	−0.62	−1.00†

†Negative self-score: motif scores below Swiss-Prot background under its own PSSM (see §1.5).

Table 3: Normalised cross-reactivity specificity matrix. Rows: proteases (PSSM used for scoring). Columns: motifs being scored. Diagonal = self-score reference (±1.00). **Bold red**: ≥ 0.20 (flagged). 23 of 30 off-diagonal pairs are orthogonal (< 0.05).

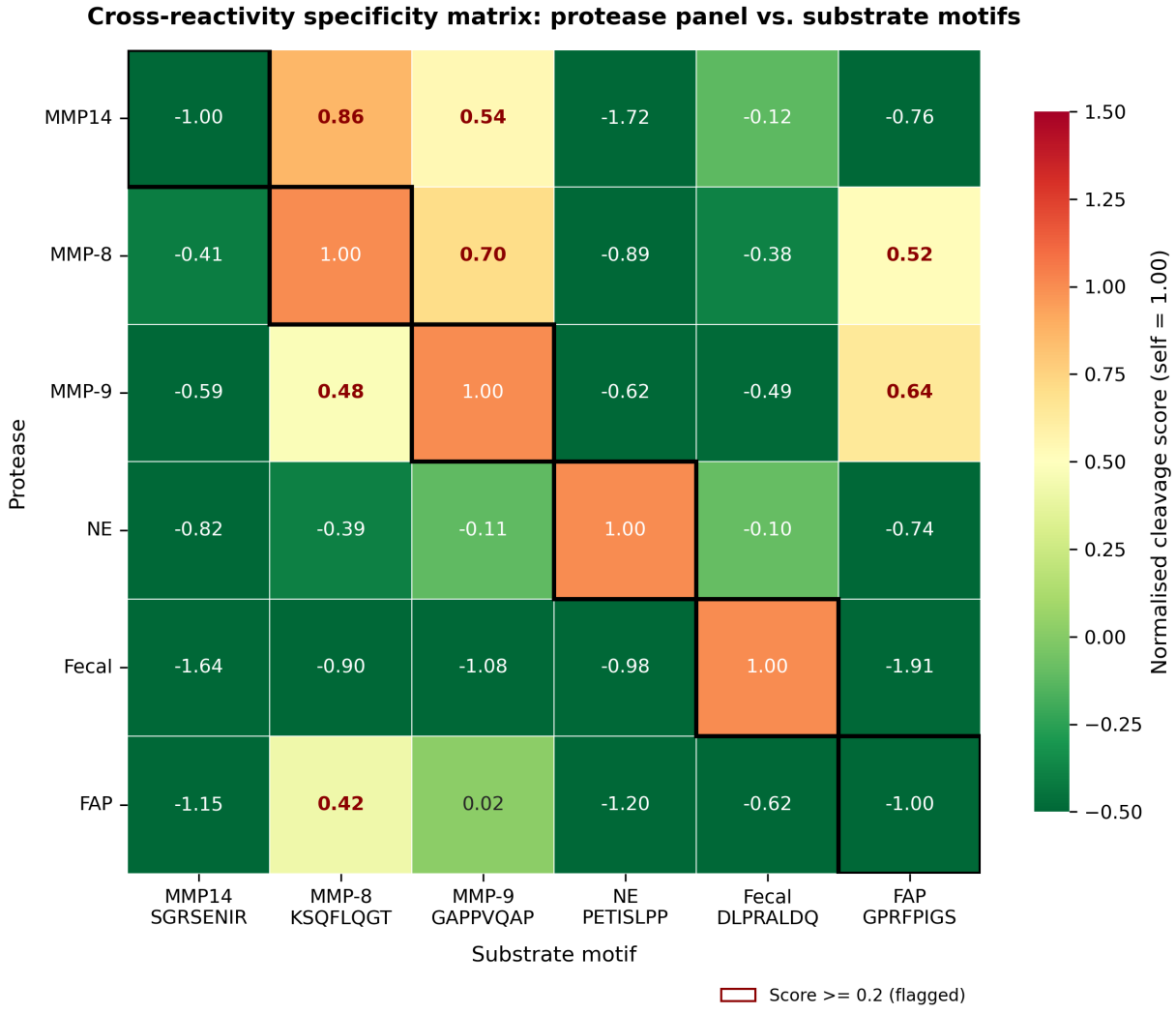


Figure 1: Specificity matrix heatmap. Rows: proteases; columns: substrate motifs. Colour scale: `RdYlGn_r` (red: cross-reactive, green: orthogonal). Diagonal cells are outlined in black. Off-diagonal values ≥ 0.20 are annotated in bold red. Panel motifs are indicated in column headers.

Flagged Pairs

Seven of 30 off-diagonal pairs exceed the 0.20 threshold. They fall into two distinct groups.

Intra-MMP cross-reactivity (five pairs). The pairs MMP14×KSQFLQGT (0.86), MMP14×GAPPVQAP (0.54), MMP-8×GAPPVQAP (0.70), MMP-9×KSQFLQGT (0.48), and MMP-8×FAP-motif (0.52) all reflect the shared M10A active-site architecture. All three gelatinolytic/collagenolytic MMPs use a Zn^{2+} catalytic mechanism with a deep hydrophobic S1' pocket accepting Phe, Leu, and Ile, and a broad S2 subsite. The ML-predicted motifs KSQFLQGT and GAPPVQAP were optimised for cleavage probability by MMP-8 and MMP-9 respectively; because the M10A pocket geometry is shared, both motifs are also well-tolerated by the other panel MMPs. This is an intrinsic constraint of the MMP subfamily: motifs that maximise individual cleavage efficiency will inevitably retain some cross-reactivity with family members. Note that the MMP-8 and MMP-9 ML selectivity indices ($S = 2.02$ and 2.39) already reflect this; they are real but moderate.

FAP-related artefacts (two pairs). MMP-9×GPRFPIGS (0.64) and FAP×KSQFLQGT (0.42) involve the FAP PSSM, which is built from 2 entries and has no statistical power. These values are not interpretable as genuine cross-reactivity and are excluded from any sensor design recommendation.

Negative Self-Scores and MMP14 SGRSENIR

MMP14 (SGRSENIR, raw self-score = −3.81) and FAP (GPRFPIGS, raw self-score = −3.47) have negative diagonal values, meaning the motif scores below the Swiss-Prot background under its own protease's PSSM. For MMP14, this reflects a documented divergence between natural ECM substrates in MEROPS (collagen, fibronectin: enriched for Pro, Gly, and Leu at P1') and the SGRSENIR biosensor motif (derived from the proMMP-2 autolytic site, enriched for Ser and Arg). SGRSENIR is experimentally validated as an MMP14 substrate ($k_{\text{cat}}/K_M = 5.5 \times 10^4$ $\text{M}^{-1}\text{s}^{-1}$; Ohuchi et al. 1997) and has been deployed in working electrochemical sensors. The PSSM mismatch reflects a MEROPS ascertainment bias towards natural ECM substrates and is not a functional deficiency of the sensor. For FAP, the negative self-score is simply noise from 2 database entries. In both cases the diagonal is normalised to −1.00 and interpreted accordingly.

Summary. The serine protease block (NE, Fecal) is cleanly orthogonal to all three panel MMPs: every cross-reactivity score in the NE and Fecal rows is negative (below Swiss-Prot background), and the MMP rows score NE and Fecal motifs at ≤ -0.10. This is the strongest structural result: the protease class boundary between serine and metalloprotease families is sufficient to ensure selectivity without motif optimisation. Within the M10 metalloproteinase subfamily, all three MMPs show mutual cross-reactivity (0.48–0.86) with the ML-predicted motifs. The NE candidate (`PETISLPP`) achieves an ML selectivity of 17.9, making it the highest-confidence motif in the panel. FAP requires either additional substrate data or a structure-based approach before a reliable specificity assessment is possible.